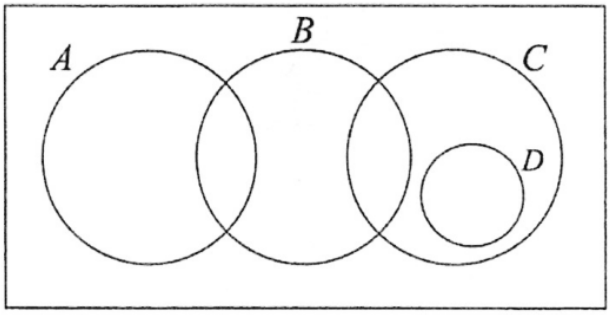
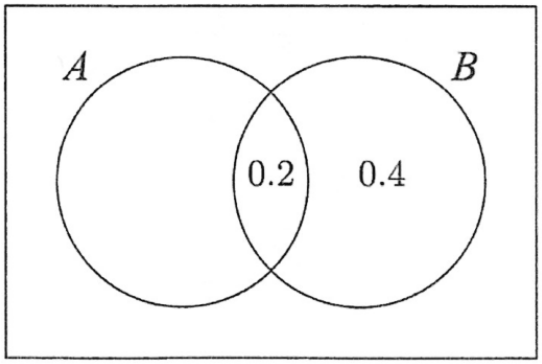
Probability: Independence and Mutually Exclusive

Questions 1–20

Question 1. Events A, B, C and D are displayed in the Venn diagram below. Write down three different examples of two events that are mutually exclusive.



Question 2. It is given that A and B are independent events. Copy and complete the following Venn diagram.



Question 3. It is given that A and B are two events such that $\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.55$ and $\mathrm{P}(A \cup B) = 0.95$. Determine if A and B are mutually exclusive.

Question 4. It is given that A and B are two events such that $\mathrm{P}(A) = 0.32$, $\mathrm{P}(B) = 0.46$ and $\mathrm{P}(A \cup B) = 0.7$. Determine if A and B are mutually exclusive.

Question 5. It is given that A and B are two events such that $\mathrm{P}(A) = \frac{3}{7}$, $\mathrm{P}(B) = \frac{2}{3}$ and $\mathrm{P}(A \cap B) = \frac{2}{7}$. Determine if A and B are independent events.

Question 6. It is given that A and B are two events such that $\mathrm{P}(A) = \frac{3}{10}$, $\mathrm{P}(B) = \frac{5}{6}$ and $\mathrm{P}(A \cap B) = \frac{1}{2}$. Determine if A and B are independent events.

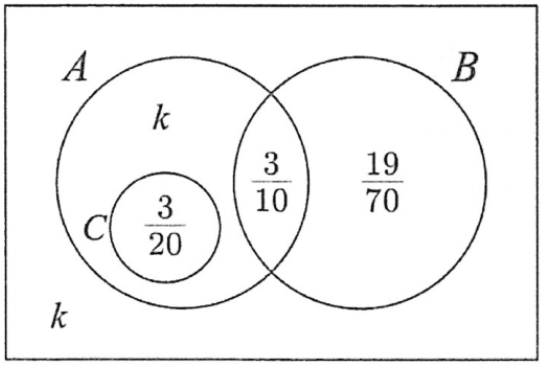
Question 7. It is given that A and B are two events such that $\mathrm{P}(A) = 0.45$, $\mathrm{P}(B) = 0.52$ and $\mathrm{P}(A \cap B) = 0.23$. Determine if A and B are independent events.

Question 8. An estate agent has 30 flats advertised on their website. 16 have an ensuite bathroom, 8 have a balcony, and 6 have both an ensuite bathroom and a balcony. A flat is chosen at random. Determine whether the events 'the flat has an ensuite bathroom' and 'the flat has a balcony' are independent.

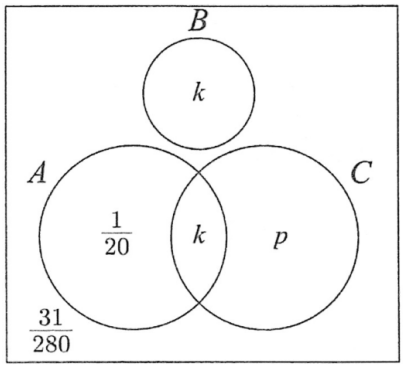
Question 9. It is given that A and B are independent events such that $\mathrm{P}(A) = 0.3$ and $\mathrm{P}(A \cup B) = 0.75$. Find $\mathrm{P}(B)$

Question 10. It is given that A and B are independent events such that $\mathrm{P}(A \cap B) = 0.85$ and $\mathrm{P}(A) = 0.95$. Find $\mathrm{P}(A \cup B)$

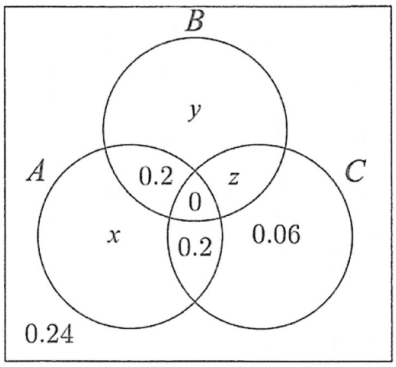
Question 11. Events A, B and C are displayed in the Venn diagram below. Determine if A and B are independent events.



Question 12. Events A, B and C are displayed in the Venn diagram below. It is given that A and C are independent events, and that A and B are mutually exclusive and B and C are also mutually exclusive. Find the value of k and the value of p



Question 13. Events A, B and C are displayed in the Venn diagram below. It is given that B and C are mutually exclusive, and that A and B are independent. Find the possible values of x, y and z



Question 14. Events A and B are independent. It is given that $\mathrm{P}(A \cap B) = \frac{1}{6}$ and $\mathrm{P}(A \cup B) = \frac{3}{4}$. Find the possible values of $\mathrm{P}(A)$ and $\mathrm{P}(B)$

Question 15. Events A, B and C have probabilities 0.05, 0.4 and 0.1 respectively.
A and C are independent events.
A and B are mutually exclusive.
B and C are also mutually exclusive. Draw a fully labelled Venn diagram showing the events A, B and C.

Question 16. Events A, B and C have probabilities 0.05, 0.1 and 0.4 respectively.
A and B are independent events.
B and C are also independent events.
A and C are mutually exclusive. Draw a fully labelled Venn diagram showing the events A, B and C.

Question 17. The two-way table below shows the events A and B. Determine whether events A and B are independent.

	A	A'
B	0.0522	0.0928
B'	0.3078	0.5472

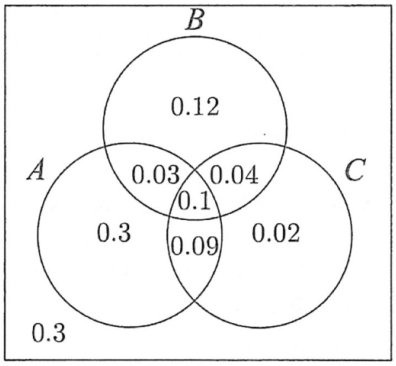
Question 18. Three classes, A, B and C in a sixth form college have 75 students, who either study three A-Levels, three A-Levels and one AS-Level, or four A-Levels. The table below shows the number and type of courses the students in each class are studying.

	Class A	Class B	Class C	Total
Three A-Levels	20	14	18	52
Three A-Levels + One AS	4	7	4	15
Four A-Levels	1	3	4	8
Total	25	24	26	75

A is the event 'the student is studying three A-Levels'.
B is the event 'the student is from Class B'.
A student is selected at random.

a) Find $\mathrm{P}(B)$
b) Find $\mathrm{P}(A \cup B)$
c) Find $\mathrm{P}(A \cap B')$
d) Determine if A and B are independent events.

Question 19. Events A, B and C are displayed in the Venn diagram below. Determine whether $A' \cap B$ and C are independent.



Question 20. Events A, B and C are such that $\mathrm{P}(A) = 0.2$, $\mathrm{P}(B) = 0.3$ and $\mathrm{P}(C) = k$. It is given that A and B are independent, and A and C are mutually exclusive. It is also given that B and C are independent and $\mathrm{P}((A \cup B \cup C)') = 0$. Find the value of k